

A Multi-Source Dataset for Kitwe City Council Financial Governance Analysis (2018–2026)

[Your Name]¹, [Your Co-author if applicable]

¹University of Zambia, School of Computing, Department of Computer Science, Lusaka, Zambia

*Corresponding author email: your.email@unza.zm

Abstract

This data article presents a multi-source dataset constructed from the digital footprints of Kitwe City Council, the largest local authority in Zambia’s Copperbelt Province. The dataset comprises 963 structured records across 18 columns extracted from 21 source documents—including approved budget estimates, audited financial statements, council meeting minutes, Constituency Development Fund (CDF) project registers, CDF empowerment grant records, and budget performance reports—spanning fiscal years 2018 to 2026. Records are organised into seven thematic sub-datasets: CDF projects (387 records), budget revenue (235 records), budget expenditure (213 records), council resolutions (88 records), financial statements (23 records), debt arrears (13 records), and empowerment grants (4 records). The dataset is grounded in Zambia’s Local Government Act No. 2 of 2019 and its subsequent amendments (Act 8 of 2023 on LGEF reform; Act 5 of 2026 on CDF allocation of K40 million per constituency). All files use pipe-separated values with the naming convention `db-unza26-csc4792-[DESCRIPTION].csv`. The dataset supports analysis of local government revenue mobilisation, CDF project implementation patterns, budget adherence, council decision-making processes, and debt management trends in Zambian local authorities.

Keywords: Kitwe City Council; Local Government Act; Constituency Development Fund; Budget estimates; Audited financial statements; Council minutes; Copperbelt Province; Zambia; Local governance data; Municipal finance

Specifications Table

Subject	Social Sciences — Public Administration and Local Governance
Specific subject area	Municipal finance, local government budgeting, CDF project implementation, council decision-making
Type of data	Structured tabular data (CSV files)
Data format	Pipe-separated values (CSV with delimiter), quoted fields
How data was acquired	Manual extraction from audited financial statements and budget performance reports; programmatic extraction (<code>pdfplumber</code> , <code>tabula-py</code> , <code>python-docx</code>) from budget PDFs, CDF project lists, and council minutes
Data source location	Kitwe City Council, Kitwe District, Copperbelt Province, Zambia (12.8°S, 28.2°E)
Data accessibility	Repository: Kaggle (https://www.kaggle.com/datasets/[username]/kitwe-city-council-dataset); Code: GitHub (https://github.com/[username]/kitwe-city-council-data)
Related research article	This is a data description article; no associated research article at time of publication

1 Value of the Data

1. **Local government financial transparency.** The dataset provides the first publicly available structured extraction of Kitwe City Council’s budget estimates, audited actuals, and financial statements spanning eight fiscal years (2018–2026). This enables longitudinal analysis of revenue mobilisation, expenditure patterns, and budget adherence in Zambia’s second-largest city.
2. **CDF implementation tracking.** With 387 records covering proposed, approved, and contracted CDF projects, the dataset offers granular visibility into how Constituency Development Fund allocations—recently increased to K40 million per constituency under Amendment Act 5 of 2026—are planned, approved, and implemented across Kitwe’s constituencies (Chimwemwe, Kamfinsa, Wusakile, Nkana, Kwacha).
3. **Council governance documentation.** The 88 council resolution, motion, and proceeding records extracted from meeting minutes provide a structured record of council decision-making, including proposer/seconder attribution and reference codes—enabling research into local democratic processes and accountability.
4. **LGEF utilisation analysis.** Revenue records from 2018–2026 explicitly track Local Government Equalisation Fund (LGEF) disbursements, enabling analysis of intergovernmental fiscal transfer effectiveness following the 2023 LGEF reform (Act 8 of 2023).
5. **Debt and arrears monitoring.** The debt arrears sub-dataset (13 records) captures outstanding obligations to utilities (ZESCO, ZAMTEL), statutory bodies (NAPSA, ZRA), and staff, supporting research into municipal debt sustainability.
6. **Reuse potential.** Researchers studying Zambian local governance, comparative municipal finance across African local authorities, CDF effectiveness, or the impact of legislative reform on local government operations can reuse this dataset. The dataset also serves as a

benchmark for data mining and information extraction from semi-structured government documents.

2 Data Description

2.1 Dataset Overview

The dataset consists of 963 structured records distributed across seven thematic CSV files and one combined file, all using pipe (|) as the field separator. The naming convention follows `db-unza26-csc4792-[DESCRIPTION].csv` where [DESCRIPTION] identifies the thematic content.

Table 1: Thematic file summary.

File Name	Rows	Record Types	Fiscal Years
db-...-cdf_projects	387	cdf_proposed (332), cdf_approved (34), cdf_project (21)	2023–2026
db-...-budget_revenue	235	budget_revenue (164), revenue (23), local_revenue (6), revenue_line (42)	2018–2026
db-...-budget_expenditure	213	budget_exp (189), expenditure (10), exp_line (14)	2018–2025
db-...-council_resolutions	88	resolution (57), motion (27), proceeding (4)	2023, 2025
db-...-financial_statements	23	annual_fs (22), audit_query (1)	2018–2024
db-...-debt_arrears	13	debt_arrears (13)	2025
db-...-empowerment_grants	4	empowerment (4)	2025
db-...-full_dataset	963	All types	2018–2026
db-...-duplicates	120	Overlapping/duplicate	Various

2.2 Column Schema

Each CSV file shares the following 18-column schema (the initial 12 columns plus 6 enrichment columns added during integration):

Table 2: Column schema for all CSV files.

Column	Type	Description
record_id	String	Unique identifier: KIT-[TYPE]-[YEAR]-[SEQ] (File 1) or KIT2-[TYPE]-[YEAR]-[SEQ] (File 2)
record_type	String	Category of the record (17 types)
fiscal_year	Integer	Zambian government fiscal year
period	String	Temporal granularity: FY2018, Q1-Q2, Jan-26, etc.
category	String	Sub-classification within the record type
line_item	String	Descriptive name of the budget line, project, or agenda item
actual_zmw	Float	Monetary amount in Zambian Kwacha; empty for budget estimates and CDF projects without amounts
description	String	Supplementary detail (contract dates, audit notes, outcomes)
source_document	String	Filename of the original PDF/DOCX source
source_table	String	Name of the table within the source document
source_page	String	Page number in the source document
stakeholders	String	Relevant parties (e.g., Kitwe City Council, Wusakile Constituency Cooperatives)
<i>Enrichment columns (added post-merge)</i>		
proposer_name	String	Name of the councillor who proposed a council resolution or motion (54 of 57 resolutions; 26 of 27 motions)
seconder_name	String	Name of the councillor who seconded a resolution or motion (54 of 57; 26 of 27)
funding_source	String	CDF funding label, e.g. “2024 CDF funding” (332 CDF proposed projects)
budget_amount_zmw	Float	Actual budget line amount in ZMW; populated for 164 revenue and 188 expenditure lines
approved	String	“Yes” for CDF approved projects (34), “No” for proposed (332); NaN otherwise
budget_year_label	String	Budget classification label, e.g. “2023 Approved Budget”, “2025 Budget Estimate” (164 revenue, 188 expenditure)

2.3 Record Types

The 17 record types are organised into seven thematic groups:

CDF Projects (387 records): CDF proposed projects (332) list community development projects submitted for Constituency Development Fund consideration; CDF approved projects (34) record projects that received formal CDF allocation; CDF projects (21) document signed/contracted projects with execution details.

Budget Revenue (235 records): Budget revenue (164) contains approved revenue estimates; revenue (23) records audited actual revenue figures; local revenue (6) captures locally generated revenue sources; revenue line annual (42) provides annual revenue line-item performance data.

Budget Expenditure (213 records): Budget expenditure (189) contains approved expenditure estimates; expenditure (10) records audited actual expenditure; expenditure line annual (14) provides annual expenditure line-item performance data.

Council Resolutions (88 records): Council resolutions (57) capture formally adopted decisions; council motions (27) record proposed but potentially unresolved items; council proceedings (4) document meeting agenda items.

Financial Statements (23 records): Annual financial statements (22) summarise audited cash receipts and payments; audit queries (1) flag outstanding audit findings.

Debt Arrears (13 records): Outstanding debt obligations to ZESCO, ZAMTEL, NAPSA, ZRA, and staff.

Empowerment Grants (4 records): CDF empowerment grant allocations to cooperatives across Kitwe’s five constituencies.

2.4 Fiscal Year Coverage

Table 3: Record distribution by fiscal year.

Fiscal Year	Records	Key Sources
2018	9	Kitwe-City-Council-2018-FS.pdf
2019	9	Kitwe-City-Council-2019-FS.pdf
2020	9	Kitwe-City-Council-2020-FS.pdf
2021	9	Kitwe-City-Council-2021-FS.pdf
2022	60	Kitwe-C-C-2022-Financial-Statements-SIGNED.pdf, Budget-Estimates-2023.pdf
2023	158	Financial statements, Budget-Estimates-2023.pdf, Council minutes (3 files)
2024	448	CDF-2024-PROPOSED-AND-APPROVED-PROJECTS.pdf, Financial statements
2025	207	Approved-Budget-2025.pdf, Budget-Performance-Report-116-LAs.pdf, Council minutes
2026	54	2026-budget-Kitwe-City-Council-amended-Final.pdf, CDF signing records

3 Methods

3.1 Data Acquisition

Source documents were obtained from two primary channels:

1. **Kitwe City Council official website** (<https://www.kitwecitycouncil.gov.zm>): Budget estimates, financial statements, and council news were downloaded directly.

2. **Ministry of Local Government and Rural Development** portal: Budget performance reports and standardised financial statement templates were obtained from the Ministry’s public document repository.

A total of 21 source documents were collected, comprising 14 PDFs, 3 DOCX files, 3 images/semi-structured documents, and 1 web page. The documents span fiscal years 2018 to 2026.

3.2 Data Extraction

Two complementary extraction approaches were employed:

File 1 — Manual curation from audited sources. The first source file (`kitwe_city_council_data__1__(1)`, 157 rows $\times$ 18 columns) was manually constructed by reading audited financial statements (2018–2024) and the Budget Performance Report for 116 Local Authorities. Manual extraction was necessary because audited financial statements use non-standard table layouts that resisted automated parsing—tables span multiple pages, use merged cells, and employ inconsistent column ordering across years.

File 2 — Programmatic data mining. The second source file (`DataMining-project.csv`, 807 rows $\times$ 9 columns) was extracted programmatically using Python libraries:

- `pdfplumber` and `tabula-py` for table extraction from budget estimate PDFs and CDF project lists
- `python-docx` for structured text extraction from council minutes DOCX files
- Regular expressions for parsing reference codes, fiscal year labels, and monetary amounts

3.3 Data Cleaning

The cleaning pipeline involved 20 documented steps organised into three phases:

Phase 1: File 1 cleaning (Steps 1–7)

1. Detected and corrected column misalignment caused by an undocumented `description` field in File 1 that shifted `source_document` through `scope` rightward by one column.
2. Removed one empty duplicate row (KIT-EXP-2023-004) where all financial values were NaN.
3. Filled missing `line_item` values by copying from the `category` column.
4. Fixed a row where `source_document` = ‘27’ was a leaked page number; restored the correct source document.
5. Fixed a row where `source_page` = ‘10/09/2026’ was a leaked date; restored page number and appended the date to the description.
6. Converted `budget_zmw` and `actual_zmw` to numeric types.
7. Stripped whitespace from all text columns.

Phase 2: File 2 cleaning (Steps 8–14)

8. Classified each row into one of three semantic types based on the source document filename.
9. Remapped misaligned columns: budget rows had amounts in `Seconder` and year labels in `Amount (K)`; CDF rows had funding source text in `Seconder`.
10. Fixed leaked source document filenames in the `Notes` column.
11. Fixed leaked budget document text in the `Amount (K)` column.

12. Extracted fiscal year from three sources depending on row type: budget year label, CDF funding source text, or council minutes filename.
13. Mapped File 2’s **Category** / **Type** values to the unified record type vocabulary.
14. Standardised the disabilities stakeholders meeting filename across inconsistent references.

Phase 3: Post-merge fixes (Steps 15–20)

15. Populated missing fiscal years for council proceedings (reference codes BPM/02/05/25 indicate 2025).
16. Corrected `source_document` = ‘Final.pdf’ to the full filename.
17. Assigned fiscal year 2024 to CDF approved projects.
18. Replaced all remaining NaN values with empty strings for clean CSV output.
19. Converted fiscal year to integer where present.
20. Removed any completely empty rows.

3.4 Data Integration

The two source files were merged into a unified 26-column superset schema, with empty placeholders for columns not present in each file. After integration, 14 columns with minimal data were dropped, yielding the final 12-column schema, to which six enrichment columns were subsequently added (see Section 3.5).

3.5 Deduplication

Three types of duplicate/overlapping records were identified and catalogued in a separate file (`db-unza26-csc4792-duplicates.csv`, 120 records):

1. **Empty duplicate row** (1 record): KIT-EXP-2023-004 with all financial values as NaN—removed from the combined dataset.
2. **Cross-file overlaps** (103 records): Revenue line items appearing in both files with different monetary values—File 1 reports audited actuals while File 2 reports budget estimates. Retained because they are semantically distinct.
3. **Within-file duplicates** (16 records): 8 pairs of rows with identical source document, subject, and seconder values. First occurrences retained; duplicates moved to the duplicates file.

3.6 Enrichment Merge

An additional raw source file (`kitwe_combined_dataset.csv`, 1770 rows × 32 columns) contained supplementary fields not captured in the initial extraction pipeline. Key observations about the raw enrichment data:

1. The **Proposer** column contains councillor names for council resolutions and motions (80 non-null rows).
2. The **Seconder** column has **mixed semantics**: councillor names (80 rows), CDF funding labels such as “2024 CDF funding” (332 rows), approval flags “Approved project” (33 rows), numeric budget amounts (340 rows), and zeros (13 rows).
3. The **Amount (K)** column is misleadingly named—it contains budget *year labels* (e.g., “2023 Approved Budget”, “2025 Budget Estimate”), not monetary amounts. The actual

monetary amounts for budget lines are stored in the **Seconder** column, indicating a column misalignment in the original extraction.

A classification function was applied to **Seconder** to decompose it into four distinct enrichment fields (**seconder_name**, **funding_source**, **budget_amount_zmw**, **approved**). The **Amount (K)** column was mapped to **budget_year_label** and **Proposer** to **proposer_name**.

Positional matching within duplicate **match_key** groups (concatenation of source document and line item, case-insensitive) was required because CDF projects and council motions often share identical descriptions (e.g., eight rows of “Construction of Wall Fence”). Two duplicate rows arising from budget misalignment (KIT2-BR-2026-216 and KIT2-CA-2024-807) were deduplicated, retaining the row with populated data.

Post-merge corrections included setting **approved** = ‘Yes’ for all CDF approved project records and **approved** = ‘No’ for all CDF proposed project records with a non-null **funding_source**.

3.7 Thematic Splitting

The 963-record combined dataset was partitioned into seven thematic CSV files based on record type mappings.

All files use pipe (|) as the field separator and are named following the convention **db-unza26-csc4792-[DESCRIPTION]**.

4 Limitations

1. **Temporal gaps:** Financial statement records for 2018–2021 contain only summary-level receipts and payments (9 records per year), whereas years 2022–2024 have more granular line-item data. This limits comparability across all fiscal years.
2. **Missing budget actuals:** The **actual_zmw** column is populated primarily for records from audited financial statements (File 1) and is empty for budget estimates (File 2). Comparative analysis of budget vs. actual requires joining across the two source types.
3. **Council minutes coverage:** Only three council meeting minutes documents (October 2023, April 2023, June 2023) and one stakeholders meeting were available digitally. Council decisions from other periods are not represented.
4. **Debt arrears point-in-time:** The debt arrears sub-dataset reflects obligations at a single point in time (Q1–Q2 2025) and does not track arrears trajectories over time.
5. **Monetary precision:** All ZMW amounts are stored as floating-point numbers, which may introduce rounding artifacts for very large figures.
6. **Single-council scope:** The dataset covers only Kitwe City Council and cannot be generalised to all Zambian local authorities without additional data collection.

5 Ethical Considerations

All source documents used in this dataset are publicly available from the Kitwe City Council website and the Ministry of Local Government and Rural Development portal. No confidential, personal, or restricted-access data was used. Council minutes contain the names of elected councillors and council officers in their official capacities; these are matters of public record per Zambia’s Constitution and the Local Government Act. The **proposer_name** and **seconder_name**

enrichment columns derive directly from these public records. No personal data of private citizens is included.

CRedit Author Statement

- **[Your Name]:** Conceptualisation, Methodology, Software, Data curation, Writing — original draft, Writing — review & editing, Visualisation
- **[Co-author if applicable]:** Supervision, Writing — review & editing

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The dataset is publicly available on Kaggle: [https://www.kaggle.com/datasets/\[username\]/kitwe-city-council-dataset](https://www.kaggle.com/datasets/[username]/kitwe-city-council-dataset)

The source code for data extraction, cleaning, and integration is available on GitHub: [https://github.com/\[username\]/kitwe-city-council-data](https://github.com/[username]/kitwe-city-council-data)

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